Umang Goel

623-286-8901 | goelumangjob@gmail.com | LinkedIn

SUMMARY

Software Engineer with strong foundations in data structures, algorithms, concurrency, and systems programming. Proficient in **C++,Java** experienced optimizing performance-critical components through profiling, memory analysis, and concurrency control. Demonstrated ability to improve latency and memory efficiency in distributed systems. Passionate about building high-performance, scalable storage and data systems.

EDUCATION

Arizona State University **Aug 2022 - May 2026**
Bachelor's, Computer Science *Tempe, AZ*

- **GPA:** 3.8/4.0
- **Coursework:** Data Structures & Algorithms, Software Engineering, Distributed Systems, Software Analysis & Design, Cloud Engineering, Object-Oriented Programming

TECHNICAL SKILLS

- **Languages**: C, C++, Java, Python
- **Backend & System**: gRPC, Flask, FastAPI, REST APIs, Distributed Systems, Concurrency, Performance Optimization, Asynchronous Processing,
- **Data and Storage**: SQL, PostgreSQL, Redis, RabbitMQ, ETL Pipelines, Data Modeling, Query Optimization, Caching Strategies
- **DevOps and Cloud** : Docker, Kubernetes (GKE), AWS, CI/CD (GitHub Actions), Linux, Scalability, Fault Tolerance, Load Balancing, System Design, TDD, Agile/Scrum

PROFESSIONAL EXPERIENCE

Honeywell and ASU Innovation Hub **Aug 2025 - Dec 2025**
Software Developer Extern *Tempe*

- Developed performance-sensitive backend components in **C++ and C#**, reducing **end-to-end latency by 25%** through architectural improvements and efficient data handling.
- Designed concurrent **ETL pipelines** with Airflow to ingest and transform **50K+ records**, managing **PostgreSQL schemas and Redis NoSQL** databases for effective database management and enabling reliable, repeatable processing for analytics and model training environments.
- Profiled compute-intensive modules using **memory and CPU profiling tools**, **reducing memory usage by 40%** and improving runtime efficiency under high-load conditions.
- Applied concurrency control and **thread-safe design patterns** to ensure correctness and stability in **multi-threaded environments.**

DMML Lab, Arizona State University **Dec 2024 - May 2025**
Software Developer Intern *Tempe*

- Designed and implemented distributed backend services in **C++ and Java**, applying **microservice and gRPC-based communication patterns** to **reduce p95 latency by 30%** across real-time workloads.
- Built high-throughput data pipelines processing **300K+ records**, applying **concurrency control, retry mechanisms, and memory-efficient data handling** to improve analytical accuracy by **15%**.
- Developed event-driven systems using **RabbitMQ with asynchronous consumers,** sustaining **500+ events/min with fault tolerance and resilient load distribution.**
- Improved system reliability and maintainability by introducing **unit testing (TDD), automated CI/CD pipelines, and performance monitoring**, reducing production defects and accelerating release cycles.

PUBLICATION

- Saketh Vishnubhatla, Alimohammad Beigi, Rui Heng Foo, **Umang Goel**, Ujun Jeong, Bohan Jiang, Adrienne Raglin, Huan Liu.An Interventional Approach to Real-Time Disaster Assessment via Causal Attribution.*ACM International Conference on Information and Knowledge Management (CIKM 2025), Demo Track.* ***https://dl.acm.org/doi/10.1145/3746252.3761487***

PROJECTS

Real-Time Event Processing Engine **2025**

- Designed and implemented a **high-performance event processing system** capable of sustaining **10K+ events/sec** with **sub-50ms latency**.
- Implemented **worker coordination, retry logic, and memory-efficient data structures** to maintain correctness under load.

Causal MMD **2025**

- Designed scalable **Flask application integrating 1.03M+ records** with ensemble classifiers, achieving **92% accuracy** for real-time disaster response coordination
- Applied performance optimization through **async I/O and batch processing** with **connection pooling, scaling throughput to 10K+ points/min** while **reducing latency by 30%**.
- Built thread-safe **REST API** with versioned endpoints and OpenAPI/Claude docs, containerized with **Docker and deployed via CI/CD** with **95%+ test coverage.**